

Risk-budgeted reversible adaptation for compact segmentation

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Research pillars: Continuous Adaptation, Security & Reliability; Scalable, Explainable & Interactive AI.

Motivation and significance

A segmentation model deployed on a constrained device must cope with changing illumination, acquisition settings, and image quality. A field robot may encounter shadows or blur; a medical imaging system may encounter a different camera response. A compact network can reduce storage and computation, yet those savings do not establish that its predictions remain trustworthy when the input distribution changes.

I propose a commit controller that estimates whether an update will help before it changes the deployed model. The update first runs on a shadow copy. Boundary and multi-view consistency measurements are then used to calculate a validation-calibrated lower bound on the expected Dice change. The candidate is committed only when that bound is positive and the clean-retention and latency budgets still have room. Every other candidate is discarded, leaving the frozen model unchanged.

Central question and hypothesis

The main question is whether a calibrated lower bound on update benefit can reject harmful test-time changes without blocking the useful ones. I expect decisions based on predicted Dice change to be safer than decisions based only on confidence or entropy, while still meeting fixed accuracy-retention and latency limits.

Scope and practical value

The initial study concerns binary segmentation with MK-UNet and accessible RGB datasets. Medical and agricultural experiments are separate tasks with separate trained models. No claim is made that the semantics of a polyp and a plant are interchangeable. The common object of study is the behavior of the adaptation mechanism, including its failure modes and computational cost.

The practical result will be a small, auditable prototype that can run on the available laptop GPU. Clinical use and autonomous crop treatment remain outside this study because each would require independent validation in its operating environment.

Problem definition and research gap

Formal setting

Let $f(x; \theta, \phi)$ output a foreground probability at each pixel. Source training learns backbone weights θ and a small adaptable parameter subset ϕ . At deployment, an unlabeled target image x arrives from a shifted distribution. The objective is to improve segmentation risk on that target distribution while respecting a latency budget and retaining source-domain behavior. Target labels are reserved for offline assessment.

Related work

MK-UNet [1] supplies the compact backbone. Tent [3], EATA [4], and SAR [5] establish entropy adaptation, sample selection, and stability controls. GraTa [6] aligns update gradients; A3-TTA [7] uses anchors and anti-forgetting; TEGDA [8] estimates test-time quality for medical segmentation. These methods make a simple uncertainty gate insufficient as a novelty claim.

The method differs in what it asks before an update is accepted: will this candidate outperform the frozen prediction? It estimates that difference directly, places a calibrated lower bound around it, and connects the decision to a cumulative risk budget and exact rollback. Calibration uses only source-validation images under shifts fixed in advance. Experiments must still show whether this combination provides a real advantage.

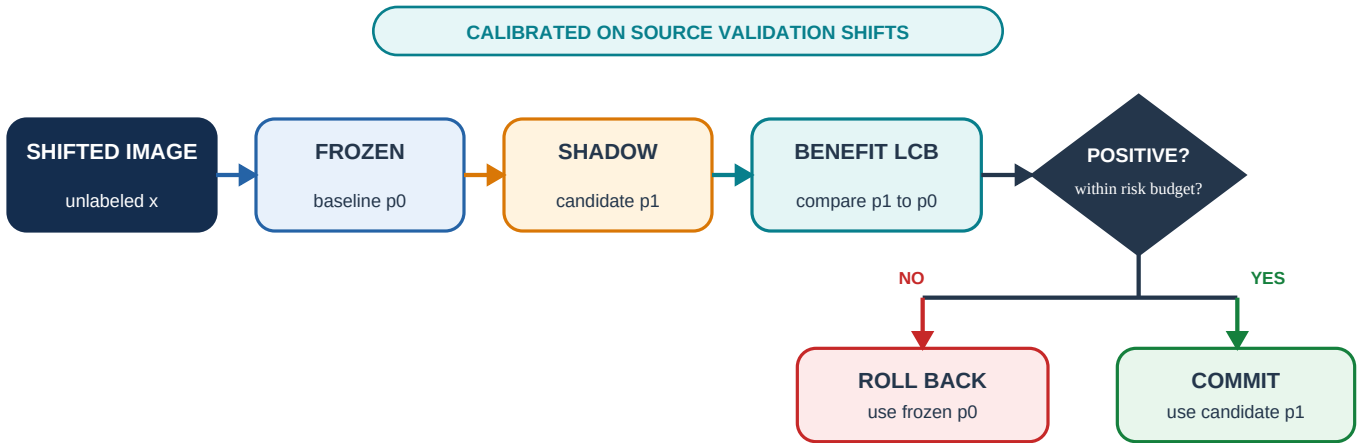
Failure cases to make measurable

Three cases will be tracked explicitly: nearly empty predictions that minimize entropy without recovering the object; localized boundary errors that are hidden by a good mean image score; and drift after a long run of similar or corrupted frames. A fourth practical failure is exceeding the inference budget because adaptation requires repeated forward and backward passes.

Research objectives

The work has three objectives: measure how the frozen backbone fails under controlled shifts; test whether calibrated decisions reject harmful updates; and find the lowest update frequency that preserves any adaptation gain. Each experiment includes the same frozen-model control, a matching ablation, and a failure rule chosen before the final test run.

Calibrated commit controller



A rejected candidate leaves the deployed model unchanged.

Candidate update and benefit features

For each image, retain frozen prediction p_0 and take one entropy-consistency step on a shadow copy of normalization affine parameters. Compute candidate prediction p_1 . Features summarize changes in cross-view disagreement, boundary stability, foreground area, entropy, and update norm. A lightweight regressor predicts $\Delta\text{Dice} = \text{Dice}(p_1, y) - \text{Dice}(p_0, y)$; target labels are never inputs.

Validation calibration

Fit the regressor on one portion of source-validation corruptions and use a disjoint calibration portion to estimate a one-sided residual quantile. The commit score is the predicted benefit minus this margin. Commit only when this lower confidence bound is positive, latency is within budget, and the cumulative clean-retention budget is unspent. Otherwise restore the exact frozen snapshot and return p_0 .

Guarantee boundary and audit trail

Split-conformal coverage relies on exchangeability between calibration and deployment shifts. It is not guaranteed under an unseen hospital or field distribution. The study therefore reports calibration coverage separately on synthetic validation shifts and on sealed real external data. Every candidate logs its features, bound, decision, latency, parameter hash, and rollback result.

A tested prototype in the repository implements the conformal lower-bound calibrator, risk-budget decision, and rollback semantics. Training integration and empirical evaluation remain planned research, keeping completed evidence separate from proposed claims.

Experimental design and evaluation

Data and separation

All required data are available locally and reproducible from public sources. ClinicDB provides 489/61/62 source train/validation/test images. CVC-ColonDB provides 380 images sealed for external medical testing; its labels cannot tune the model, calibrator, thresholds, or stopping rules. CWFID provides a separate 31/8/21 agricultural feasibility branch after correcting published train/test overlap. Manifests record every file hash and confirm zero exact-image overlap between ClinicDB and ColonDB.

Controls and ablations

Compare the frozen model, normalization-affine entropy adaptation, augmentation averaging, and the proposed controller. Ablate the benefit model, conformal margin, cumulative budget, and rollback. Use identical checkpoints, image order, preprocessing, and thresholds. A random-commit control tests whether selection adds value beyond updating at the same frequency.

Shift suite

Start with preregistered brightness reduction, blur, and additive noise, evaluated independently and in sequences that alternate clean and shifted inputs. Run both episodic adaptation, which resets after each image, and continual adaptation, which retains accepted parameters. Corruption severity is controlled and recorded; synthetic corruption is a diagnostic proxy, not a substitute for real scanner or field shifts.

Outcomes and statistics

Primary outcomes are mean per-image Dice and the paired Dice change relative to the frozen model under each shift. Report IoU, boundary-sensitive errors, update acceptance rate, clean-domain retention, peak GPU memory, and median and 95th-percentile end-to-end latency. Use at least five training seeds and report seed variability separately from image bootstrap uncertainty. Where grouping metadata exist, resample acquisition groups rather than individual frames.

Decision rule

A planned feasibility target is a positive paired improvement on shifted data with no more than a one-percentage-point mean Dice loss on clean data. This is a research target, not a measured result. Report every prespecified condition, including failures, and separate method selection on validation data from the final test comparison.

Preliminary evidence and expected outcomes

Completed backbone experiments

Experiment	Test n	Dice / IoU	Best epoch
ClinicDB	62	91.28% / 85.19%	200
CWFID	21	87.98% / 78.59%	128

The ClinicDB run achieved 91.28% Dice and 85.19% IoU. Relative to the published 93.48% Dice reference [1], the difference is -2.20 percentage points. This comparison is descriptive: the publication averages five runs, whereas this study uses one seed and corrects evaluation and resizing behavior in the current public code.

On CWFID, the independently trained model achieved 87.98% Dice and 78.59% IoU on 21 held-out images. The target is all vegetation, merging crops and weeds. These results demonstrate that the architecture can be trained on an agricultural task; they do not establish zero-shot medical-to-agricultural transfer or robustness across farms.

How the evidence supports feasibility

The preliminary runs verify that this laptop can train and evaluate the selected architecture on both tasks and that the data pipeline can preserve traceable split and checkpoint identities. They support the feasibility of a modest adaptation study. They do not test the proposed gate, demonstrate a causal benefit from adaptation, or establish that the research hypothesis is true.

Expected scientific outputs

The study will produce an adaptation benchmark, working rollback code, an accuracy-versus-cost comparison, and spatial examples of updates that helped or failed. If the frozen model or augmentation averaging performs better within the compute budget, that result will be reported directly. Settings are frozen before the sealed test evaluation, so the test set cannot become another tuning set.

Limits and mitigation

The largest risks are calibration-set size, non-exchangeable real shifts, correlated frames, and threshold sensitivity. Report bound coverage, selective risk versus commit rate, and failures on sealed ColonDB in addition to mean accuracy. Multiple seeds and a frozen protocol limit researcher degrees of freedom. Raw probabilities are required for calibration; replication-only min-max normalization is unsuitable for reliability estimation.

Work plan and reproducibility

Period	Milestone and decision
Week 1	Repeat baseline seeds; integrate the shadow update; lock shifts and data partitions.
Week 2	Fit the benefit model and calibrator; freeze settings; run ablations and latency tests.
Week 3	Evaluate sealed ColonDB and CWFID branches; stress test streams; analyze and write.

Reproducibility and resources

The project retains upstream provenance, data manifests and hashes, training histories, checkpoint identities, and per-image metrics. ClinicDB, ColonDB, and CWFID are acquired; raw files remain outside Git under their original terms. The adaptation study will add calibration folds, update decisions, bounds, rollback checks, and timing records. The one-step shadow design is implementable on the available RTX 5070 laptop.

The proposal describes planned research. Only the baseline results on page 5 are completed experiments; all adaptation results remain future work.

References

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